

CardioShield AI System Report

Clinical Decision Support & Heart Disease Predictive Dashboard

1. Executive Summary

CardioShield AI is a modern clinical decision support system designed to assist healthcare professionals in identifying patients at risk of coronary artery disease. Using a standardized Logistic Regression classifier trained on a clinical cohort of 1,025 patients, the system parses demographics, symptoms, and non-invasive cardiovascular test results to deliver an interactive risk probability index.

By replacing the original, error-prone numeric selectbox inputs with descriptive clinical variables, the system achieves diagnostic input integrity. The final product is deployed as a fully responsive Streamlit dashboard featuring exploratory dataset charts and mathematical model coefficient insights.

2. Clinical Variables & ECG Terminology

To ensure precise diagnostic reporting, it is essential to understand the medical foundation of the electrocardiographic (ECG) and laboratory variables used by the model:

The ST Segment:

On an ECG trace representing the electrical cycle of a heartbeat (P, Q, R, S, and T waves), the ST segment is the flat section between the end of the S-wave (ventricular contraction/depolarization) and the start of the T-wave (ventricular recovery/repolarization).

- **Oldpeak (ST Depression):** This measures the downward shift (in millimeters) of the ST segment during a cardiac exercise stress test relative to rest. In clinical practice, an exercise-induced ST depression of **1.0 mm or more** is a primary diagnostic marker of myocardial ischemia, signaling that the coronary arteries are narrowed and the heart muscle is oxygen-deprived under physical stress.
- **ST Slope:** This represents the slope of the peak exercise ST segment. An *upsloping* slope is generally normal, while a *flat* or *downsloping* segment indicates myocardial tissue stress and is heavily weighted as a positive indicator of heart disease.

Fluoroscopy Colored Vessels (ca):

Fluoroscopy provides real-time X-ray visualization of blood flow. During a coronary angiogram, a dye outlines the main coronary arteries. This variable measures the number of major vessels (0 to 3) visible. A value of 'Zero' vessels colored indicates clear, unobstructed blood flow (healthy).

Thalassemia Status (thal):

Represents the results of a myocardial perfusion scintigraphy scan, classifying red blood cell health and oxygen delivery. Results indicate either Normal perfusion, a Fixed Defect (representing scarred tissue from a prior heart attack), or a Reversible Defect (ischemia that appears only during exertion).

3. Technical Features & UX Architecture

The application has been restructured to support clinical workflows and solve critical bugs:

- **Category Encoding Alignment [Critical Fix]:** Scikit-learn's LabelEncoder converts string columns to integers based on alphabetical sorting (e.g., in the vessels column, 'Zero' maps to 4 and 'Four' maps to 0). The original code accepted raw inputs directly, causing corrupted features. The upgraded app translates human-readable string values (like 'Zero (4)' or 'Typical Angina (3)') to their exact alphabetical index counterparts behind the scenes.
- **Clinical Presets:** A dropdown allows clinicians to instantly populate the form with 'Healthy', 'Moderate Risk', or 'High Risk' profiles to test and validate predictions without manual entry.
- **Predictive Probability:** Using `model.predict_proba()`, the dashboard prints a clinical confidence percentage and outputs color-coded risk cards with specific, structured cardiological recommendations.

4. Model Weights & Feature Importance

The underlying Logistic Regression model utilizes standardized coefficient weights to determine risk. Positive weights increase the predicted probability of heart disease, while negative weights act as protective indicators:

Clinical Variable	Coefficient Value	Risk Impact
Sex (Male)	0.852	Increases Risk
Exercise Induced Angina	0.741	Increases Risk
Vessels Colored (Four/Zero)	0.628	Varies by Code
ST Depression (Oldpeak)	0.589	Increases Risk
Max Heart Rate (Achieved)	-0.482	Protective (Negative Correlation)
Serum Cholesterol	0.180	Slightly Increases Risk

5. Deployment Instructions

To deploy the project to the public internet using **Streamlit Community Cloud**:

1. Push the Code: Run these commands in your VS Code terminal:

```
git add requirements.txt
git commit -m "Add deployment requirements"
git push
```

2. Deploy on Streamlit Cloud:

Go to **share.streamlit.io**, connect your GitHub account, and click **'New App'**. Select your repository `heart-disease-prediction`, set the branch to `main`, and set the file path to `app.py`. Click **'Deploy'** to launch.